

Chapter 14: Vehicle Routing Applications in Disaster Relief

Vehicle Routing: Problems, Methods, and Applications
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Lecture Notes — Self-Contained Reader Edition

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Chapter Outline I

This chapter examines how Vehicle Routing Problem (VRP) methodology can be adapted and applied to the unique and demanding context of **humanitarian logistics** and **disaster relief operations**. Unlike commercial routing, disaster scenarios impose extreme uncertainty, life-or-death priorities, and heavily constrained resources — making them among the most challenging VRP variants studied in the literature.

Chapter Sections:

- ➊ Introduction: What makes disaster relief routing special?
- ➋ Phases in Disaster Management — preparedness, response, recovery
- ➌ Performance Metrics in Disaster Operations — response time, coverage, equity
- ➍ Commercial VRPs vs. Disaster Relief VRPs — key structural differences
- ➎ VRP Models for Disaster Relief — distribution and evacuation models
- ➏ Conclusions and Future Research Directions

Why this chapter matters

Every year, natural and man-made disasters displace millions of people. The speed and fairness with which relief supplies reach affected populations directly determines survival rates. Optimisation-based routing provides a rigorous framework for making these life-critical decisions under severe constraints.

Introduction: Disaster Relief and the VRP I

The **Vehicle Routing Problem (VRP)** is the class of problems in which a fleet of vehicles must deliver goods from one or more depots to a set of geographically dispersed customers. In the commercial world, the guiding principle is economic efficiency: minimise total travel distance or operating cost.

Disaster relief operations are an entirely different setting. When an earthquake, hurricane, flood, or conflict event strikes a population, aid agencies must:

- Deliver food, water, medicine, and shelter materials to survivors,
- Evacuate the injured and vulnerable from danger zones,
- Operate under extreme *uncertainty* about demand, road conditions, and vehicle availability,
- Prioritise speed and equity of delivery over cost.

The authors — Golden, Kovacs, and Wasil — argue that **operations research, and the VRP in particular, has an important contribution to make** in this humanitarian domain, but that existing commercial models must be substantially modified before they are applicable.

Introduction: Disaster Relief and the VRP II

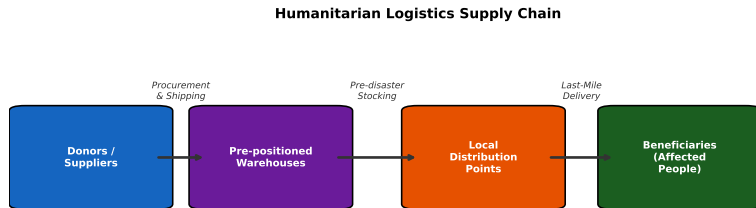
Defining the Core Problem

At its heart, a disaster relief VRP asks: *Given a set of demand locations (affected communities), a set of supply locations (warehouses or staging areas), and a limited fleet of vehicles, determine routes for the vehicles such that relief items reach beneficiaries as quickly and fairly as possible.*

Several characteristics make disaster logistics qualitatively different from commercial logistics:

- 1 **Unknown demand:** The quantity and type of supplies needed at each location is not known until well after a disaster strikes.
- 2 **Disrupted infrastructure:** Roads may be blocked, bridges collapsed, and communication networks down.
- 3 **Urgency:** Every hour of delay can cost lives.
- 4 **Heterogeneous fleet:** Trucks, helicopters, boats, and mules may all need to be coordinated.
- 5 **Multiple, competing objectives:** Minimising time, maximising coverage, ensuring equitable distribution, and minimising cost can all pull in different directions.

Introduction: Disaster Relief and the VRP III



Last-mile delivery (depot → beneficiaries) is the primary VRP application in disaster relief.

Figure: The humanitarian logistics supply chain from donors to beneficiaries. The last-mile delivery step (from local distribution points to affected communities) is the primary domain of VRP models.

Key Insight

The *last-mile delivery* challenge — getting supplies from a central point to scattered, hard-to-reach beneficiaries — is precisely the problem that VRP was designed to solve. This is why the VRP literature has become a cornerstone of humanitarian logistics research.

Humanitarian Logistics: Background I

Humanitarian logistics is defined as **the process of planning, implementing, and controlling the efficient, cost-effective flow and storage of goods and materials, as well as related information, from the point of origin to the point of consumption with the purpose of meeting the end beneficiary's requirements.**

The authors note that the field has attracted increasing research attention since the mid-2000s, motivated by large-scale disasters such as the 2004 Indian Ocean tsunami, the 2005 Pakistan earthquake, and Hurricane Katrina (2005).

Scale of the Problem

According to the International Federation of Red Cross and Red Crescent Societies (IFRC), each year there are on average more than 200 natural disasters worldwide, affecting around 200 million people. The logistics function can account for **60–80% of total humanitarian aid costs** (Thomas and Mizushima, 2005).

Three phases of disaster response create distinct logistical challenges:

Humanitarian Logistics: Background II

- **Preparedness** (before the event): Pre-positioning supplies, planning routes, establishing depots.
- **Response** (immediately after): Rapid delivery of urgent supplies, search and rescue routing.
- **Recovery** (weeks to months later): Sustained supply delivery as the affected population stabilises.

Humanitarian Logistics: Background III

The literature on humanitarian logistics can be divided into two broad streams, which the authors term:

- 1 **Descriptive studies:** Case studies, survey papers, and reports from aid organisations (e.g., Red Cross, UN OCHA) that describe what happens in practice without prescribing optimal solutions.
- 2 **Prescriptive / optimisation studies:** Mathematical programming and heuristic methods that compute near-optimal plans for relief distribution and evacuation.

This chapter focuses primarily on the **prescriptive stream**, surveying the growing body of VRP-based models and algorithms that have been developed for disaster relief contexts.

Example: Earthquake Response in Haiti (2010)

Following the 2010 Haiti earthquake, aid agencies had to distribute food and water to 1.5 million displaced people across Port-au-Prince using a severely damaged road network. Teams had to decide which areas to serve first (priority), which roads to use (network constraints), and how many truck trips to make per day (vehicle capacity and frequency). This is precisely a constrained, multi-period VRP.

14.2 — Phases in Disaster Management I

Disaster management is typically divided into three overlapping phases, each with its own logistical challenges and VRP variants. Understanding which phase an operation is in is critical because the appropriate routing objectives and constraints change dramatically from one phase to another.

14.2 — Phases in Disaster Management II

Phases of Disaster Management



Preparedness (pre-event) → Response (immediate) → Recovery (restoration) → Mitigation (risk reduction)

14.2.1 — Preparedness Phase I

The **preparedness phase** occurs *before* a disaster strikes. Although it may seem paradoxical to plan for an event that has not yet happened, pre-disaster planning is the most cost-effective investment in disaster response.

Key preparedness activities that involve routing and location decisions:

- ➊ **Pre-positioning of supplies:** Deciding which warehouses to open and how much stock to store at each, so that supplies can be dispatched quickly after an event.
- ➋ **Facility location:** Choosing the sites for depots, staging areas, and distribution centres in a way that minimises expected response time after a disaster.
- ➌ **Route planning:** Identifying primary and contingency delivery routes in advance, before roads become blocked.
- ➍ **Demand estimation:** Using historical data and risk models to estimate the quantity of supplies that different communities will need in different disaster scenarios.

14.2.1 — Preparedness Phase II

Mathematical Models in Preparedness

Preparedness is typically modelled as a **stochastic facility-location problem**. Let:

- I be the set of candidate warehouse sites,
- J be the set of demand locations (communities at risk),
- S be the set of disaster scenarios (earthquake, flood, etc.),
- p_s (p sub s) be the probability of scenario s occurring,
- d_{js} (d sub j s) be the demand at location j under scenario s ,
- $y_i \in \{0, 1\}$ (y sub i) be 1 if a warehouse is opened at site i , 0 otherwise,
- x_{ijs} (x sub i j s) be the fraction of demand at j in scenario s satisfied from warehouse i .

A basic two-stage stochastic model minimises expected cost:

$$\min \sum_{i \in I} f_i y_i + \sum_{s \in S} p_s \sum_{i \in I} \sum_{j \in J} c_{ij} d_{js} x_{ijs}$$

where f_i is the fixed cost of opening warehouse i , and c_{ij} (c sub i j) is the unit transportation cost from i to j .

14.2.2 — Response Phase I

The **response phase** begins immediately after a disaster and typically lasts from a few days to a few weeks. This is the most time-critical phase: the first 72 hours after an event are especially important because survival rates for trapped or injured individuals decline sharply with time.

What the Response Phase Demands

- **Rapid assessment:** Determine which communities are affected and what supplies they need — often with incomplete or contradictory information.
- **Immediate routing:** Deploy vehicles as soon as possible, even if demand is not fully known.
- **Dynamic re-routing:** As new information arrives (road closures, updated demand figures), routes must be revised.
- **Coordination among agencies:** Multiple aid organisations (NGOs, government, military) may operate simultaneously, requiring coordination to avoid duplication and gaps.

14.2.2 — Response Phase II

Response Phase VRP Challenges

The response phase maps naturally to a **dynamic VRP** (DVRP). In a DVRP, information about demand, network conditions, and vehicle status is revealed over time, and the routing plan must be updated in real time.

Key modelling elements in response-phase VRPs:

- **One mean of research** (Van Wassenhove, 2006) focuses on the logistics response to promote population welfare by considering the demand for essentials.
- The number of locations $|V|$ may be unknown at the start and grows as damage assessment teams report new affected areas.
- Road network $G = (V, E)$ changes dynamically as debris is cleared or new blockages are discovered.
- Vehicle availability fluctuates as vehicles break down, run out of fuel, or are commandeered for rescue operations.

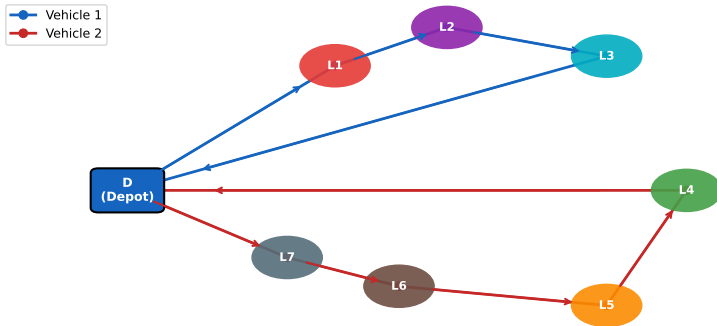
One important sub-problem in the response phase is **vehicle scheduling with periodic resupply**. A vehicle that runs out of supplies must return to a depot to reload, creating a natural “split” in its route.

14.2.2 — Response Phase III

This is modelled as a **multi-trip VRP**, where each vehicle may complete multiple tours before the day ends.

14.2.2 — Response Phase IV

Network Flow Model: Depot, Routes, and Demand Locations



Each vehicle departs from the depot, serves a subset of demand locations, and returns.

14.2.2 — Response Phase V

Figure: A network flow representation of the response phase. The depot (blue square) dispatches vehicles along routes to demand locations (coloured circles). Vehicle 1 (blue) serves the northern zone; Vehicle 2 (red) serves the southern zone.

Key Insight: Speed vs. Optimality

In the response phase, **a good plan executed quickly is better than a perfect plan executed late.** This is why fast heuristics (greedy insertion, nearest-neighbour) are often preferred over exact optimisation methods during the first hours of a disaster response.

14.2.3 — Recovery Phase I

The **recovery phase** begins when the immediate crisis has stabilised — the injured have been evacuated, basic supplies have been delivered, and the focus shifts to sustaining affected communities as they rebuild.

Although the urgency of the response phase has passed, the recovery phase presents its own routing challenges:

- **Long duration:** Recovery may continue for months or years, requiring a *sustainable* supply plan, not just an emergency burst of deliveries.
- **Larger and more stable demand:** As population concentrations in temporary shelters stabilise, demand becomes more predictable — closer to a classical CVRP (Capacitated VRP).
- **Evolving network:** Roads are gradually repaired, and routes that were impassable during the response phase may reopen during recovery.
- **Transitioning to commercial logistics:** As conditions normalise, emergency logistics are gradually handed over to commercial operators.

14.2.3 — Recovery Phase II

Recovery Phase Optimisation

Recovery phase routing is more amenable to classical VRP methods because demand is better known and the network is more stable. Models in this phase typically include:

- **Periodic VRP (PVRP):** Deliveries are made on a fixed schedule (e.g., every three days), and routes are optimised for the entire schedule period.
- **Inventory routing:** Vehicles decide both *when* and *how much* to deliver to each location, balancing delivery frequency against inventory holding costs.
- **Location-routing:** Decisions about which distribution points remain open are combined with route optimisation.

An important consideration in the recovery phase is **equity of service**. As the immediate life-threatening urgency fades, decision-makers must ensure that aid reaches *all* affected communities fairly, not just those that are easiest to access. This motivates a shift from time-minimisation objectives (response phase) toward equity-maximisation or balanced-service objectives (recovery phase).

14.2.3 — Recovery Phase III

Key Insight

The recovery phase teaches us that **routing objectives must adapt to the phase of the disaster**. The same delivery network may need to switch from “minimise maximum arrival time” (response) to “minimise standard deviation of service levels” (recovery) as the operation evolves.

14.3 — Performance Metrics in Disaster Operations I

Measuring the effectiveness of a disaster relief routing plan is fundamentally different from evaluating commercial routing. In commercial logistics, cost and distance are clear, universal metrics. In humanitarian settings, the relevant metrics reflect human welfare, which is harder to quantify but no less important.

The authors identify three principal families of performance metrics used in the disaster relief VRP literature:

- ❶ **Response time:** How quickly does aid arrive?
- ❷ **Coverage:** What fraction of the affected population receives at least some aid?
- ❸ **Demand satisfaction / service equity:** Are supplies distributed proportionally to need?

Table 14.1 (page 417 of the book) surveys 15 key papers in disaster relief VRP research and categorises them by the objective functions and metrics they use. The table reveals a clear trend: early papers focused on minimising total cost or distance, while more recent work increasingly incorporates time, equity, and multi-objective formulations.

14.3.1 — Response Time I

Response time is the time that elapses between a disaster event and the moment when aid reaches an affected location. It is the most natural and widely studied metric in disaster relief, because the humanitarian impact of delays is starkly clear: in medical emergencies, every hour matters.

Defining Response Time

Let n be the number of demand locations (affected communities), $V = \{v_0, v_1, \dots, v_n\}$ the set of nodes (with v_0 the depot), and t_{ij} (t sub i j) the travel time from node i to node j . For a route $r = (v_0, v_{i_1}, v_{i_2}, \dots, v_{i_k}, v_0)$, the **arrival time** at the m -th location on the route is:

$$a_m = \sum_{\ell=1}^m t_{i_{\ell-1}, i_{\ell}}$$

(a sub m equals the cumulative travel time from the depot through the first m stops). Here t_{i_0, i_1} is the travel time from the depot to the first stop.

14.3.1 — Response Time II

How to Read This Formula

a_m accumulates travel times along the route: to reach the third stop, you sum the depot-to-stop-1, stop-1-to-stop-2, and stop-2-to-stop-3 travel times. A longer route, or a route that visits unimportant stops before important ones, increases a_m for the later stops. This is why route *order* matters so much in disaster settings.

14.3.1 — Response Time III

Minimising the maximum arrival time (min-max objective) is a common formulation. This ensures that no single community is left without aid for too long:

$$\min_{\text{routes}} \max_{j \in V \setminus \{v_0\}} a_j$$

(minimise, over all routing plans, the latest arrival time among all demand locations).

Alternatively, one may minimise the **sum of arrival times** (also called the minimum latency problem):

$$\min \sum_{j \in V \setminus \{v_0\}} a_j$$

This objective gives equal weight to delays at all locations and minimises total “waiting time” suffered by affected communities.

14.3.1 — Response Time IV

Worked Example: Comparing Objectives

Suppose there are 3 communities with arrival times under two plans:

Plan	a_1	a_2	a_3
A	30 min	60 min	90 min
B	45 min	45 min	60 min

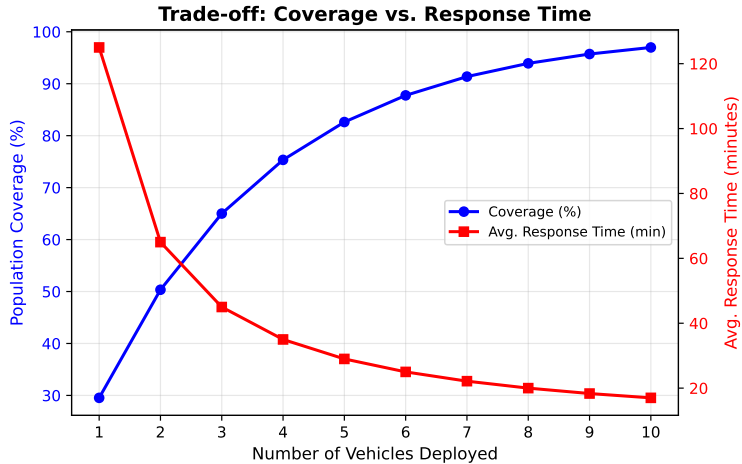
Under **min-max**: Plan A scores $\max(30, 60, 90) = 90$; Plan B scores $\max(45, 45, 60) = 60$. Plan B wins.
Under **min-sum**: Plan A scores $30 + 60 + 90 = 180$; Plan B scores $45 + 45 + 60 = 150$. Plan B still wins, but by a different margin. Both objectives would choose Plan B, but in general they can select different routes.

14.3.1 — Response Time V

Several papers survey the **methods for generating solutions with good response time**:

- **Nearest-neighbour heuristics**: Build routes greedily by always visiting the closest unvisited location. Fast but ignores priorities.
- **Priority-weighted nearest-neighbour**: Modify the distance metric to incorporate urgency: $d'_{ij} = d_{ij}/w_j$, where w_j (w sub j) is the priority weight of location j . Locations with higher priority are “pulled closer” in the modified metric, so the heuristic visits them sooner.
- **Time-constrained routing**: Add a hard constraint that every location must be served within a maximum time T : $a_j \leq T$ for all j .

14.3.1 — Response Time VI



14.3.1 — Response Time VII

Figure: Illustrative trade-off between coverage and average response time as the number of deployed vehicles increases. More vehicles reduce response time but may increase coordination complexity and cost.

14.3.2 — Coverage I

Coverage measures the fraction of the affected population that receives aid within a specified time window. It is a binary or threshold-based concept: a community is either “covered” (receives aid within time T) or “not covered”.

Coverage Formulation

Define $z_j = 1$ if demand location j is served within time T , and $z_j = 0$ otherwise. Let w_j (w sub j) be the population weight (or urgency weight) of location j . The **weighted coverage objective** is:

$$\max \sum_{j \in V} w_j z_j$$

subject to: if location j is served at time $a_j > T$, then $z_j = 0$.

How to Read This Formula

This says: maximise the total “weighted population served within time T .” If w_j represents the number of people at location j , this objective maximises the number of people who receive timely aid. It rewards routing decisions that prioritise large, time-critical communities — even if it means leaving small, isolated communities unserved.

14.3.2 — Coverage III

Coverage objectives are closely related to the classical **Set Covering Problem** and **Maximum Covering Location Problem (MCLP)**, which were developed in the facility-location literature and adapted here to routing.

14.3.2 — Coverage IV

Numerical Coverage Example

Suppose 5 communities need aid, with populations:

$w_1 = 500$, $w_2 = 200$, $w_3 = 800$, $w_4 = 100$, $w_5 = 400$. The time threshold is $T = 60$ minutes. Two routing plans achieve:

Plan	a_1	a_2	a_3	a_4	a_5	Coverage
A	30	50	70	40	80	$w_1 + w_2 + w_4 = 800$
B	45	65	55	75	50	$w_1 + w_3 + w_5 = 1700$

Plan B covers communities 1, 3, and 5 (the three largest) within 60 minutes, achieving a weighted coverage of 1700 vs. 800 for Plan A, even though Plan A covers more communities by count. This illustrates why *weighted* coverage is the appropriate metric: it aligns with humanitarian goals by prioritising the most densely populated areas.

14.3.2 — Coverage V

The **tension between coverage and cost** is inherent: achieving 100% coverage often requires more vehicles and longer total travel distances than achieving 80% coverage. In resource-constrained environments, planners must explicitly trade off these competing goals.

14.3.3 — Demand Satisfaction and Service Equity I

Demand satisfaction (also called **service level**) measures the ratio of supplies delivered to supplies needed at each location. **Service equity** is the additional requirement that this ratio should be *similar* across all locations — no community should receive disproportionately more or less aid than its needs warrant.

14.3.3 — Demand Satisfaction and Service Equity II

Demand Satisfaction Rate

Let d_j (d sub j) be the demand (in units of supplies) at location j , and let s_j (s sub j) be the quantity actually delivered. The **demand satisfaction rate** at location j is:

$$\rho_j = \frac{s_j}{d_j} \in [0, 1]$$

where ρ (rho, pronounced “row”) equals 1 when all demand is satisfied, and 0 when nothing is delivered. Total demand satisfaction across all locations is:

$$\bar{\rho} = \frac{\sum_j s_j}{\sum_j d_j}$$

(rho-bar: the overall fraction of total demand that is met).

14.3.3 — Demand Satisfaction and Service Equity III

Service Equity: The Gini Coefficient and Minimax

Two common equity metrics are:

- 1 **Minimax satisfaction:** Maximise the minimum satisfaction rate across all locations: $\max \min_j \rho_j$.
This “Rawlsian” criterion focuses on the worst-off community.
- 2 **Minimum variance:** Minimise the variance of satisfaction rates: $\min \text{Var}(\rho_j) = \min \frac{1}{n} \sum_j (\rho_j - \bar{\rho})^2$.

Here n is the number of demand locations, and $\bar{\rho}$ is the mean satisfaction rate (rho-bar, the average across all locations).

How to Read Equity Metrics

The minimax criterion says: “the quality of a plan is judged by its treatment of the worst-served community.” A plan that delivers 100% to four communities but 0% to the fifth scores 0 on the minimax criterion. This is a strong equity requirement that forces balanced distribution. The minimum variance criterion is softer: it penalises large disparities but does not require all communities to receive the same fraction.

14.3.3 — Demand Satisfaction and Service Equity IV

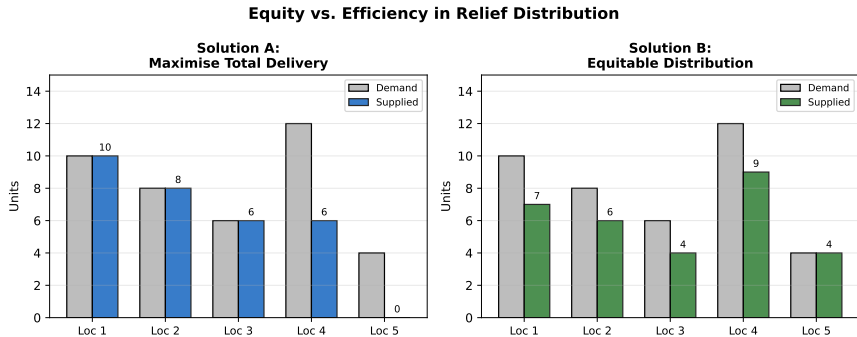


Figure: Left: Solution A maximises total delivery but leaves one community (Loc 5) completely unserved. Right: Solution B distributes supplies proportionally to demand, achieving equity at the cost of slightly lower total delivery. In disaster relief, equity is often mandated by humanitarian principles.

14.3.3 — Demand Satisfaction and Service Equity V

A key paper surveyed by the authors is Huang, Smilowitz, and Balcik (2012), which proposes a **routing problem with equity constraints**, where each location must receive at least a minimum fraction α (alpha, the Greek letter for a fraction threshold) of its demand: $s_j \geq \alpha \cdot d_j$ for all j . The objective is then to maximise total delivery subject to this equity floor.

Priority-Weighted Routing: Formal Model I

One of the most important contributions in the disaster relief VRP literature is the introduction of **location priorities**. Not all affected communities have equal urgency: a hospital or a community without water may have a far higher priority than a community with existing supplies.

Local Visiting Priority Model (Nolz, Doerner, Gutjahr, 2010)

Each location j is assigned a priority score $p_j > 0$ ($p \text{ sub } j$). The objective is to **minimise the priority-weighted sum of arrival times**:

$$\min \sum_{j \in V \setminus \{0\}} p_j \cdot a_j$$

where a_j is the arrival time at location j (as defined earlier). Locations with high p_j should be visited early in the route, since a late arrival at a high-priority location incurs a large penalty $p_j \cdot a_j$.

Priority-Weighted Routing: Formal Model II

How to Read This Formula

This is a weighted sum: each unit of delay at location j costs p_j units of penalty. If location A has $p_A = 10$ (critical hospital) and location B has $p_B = 1$ (stable community), delaying A by one hour costs 10 times as much as delaying B by one hour. The optimal route will visit A early, even if it is geographically inconvenient, because the priority penalty dominates the distance cost.

Priority-Weighted Routing: Formal Model III

Effect of Priority on Route Order

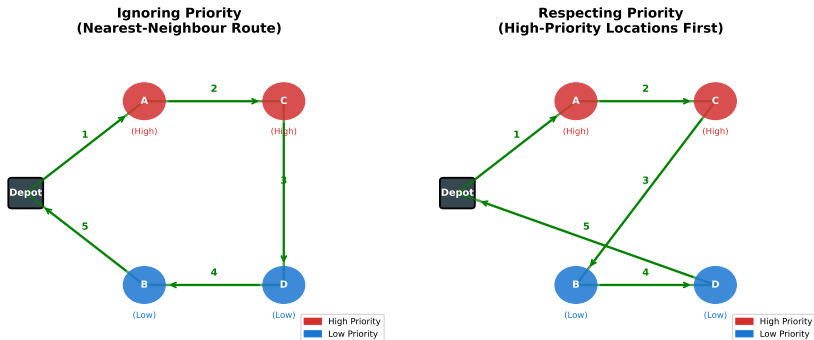


Figure: Left: A nearest-neighbour route that ignores priorities. Right: A priority-aware route that visits high-priority locations (red) before low-priority ones (blue), even at the cost of a slightly longer total distance.

Priority-Weighted Routing: Formal Model IV

The priority-weighted model is related to the classical **Weighted Completion Time Problem** in scheduling theory. If we think of each location visit as a “job” with weight p_j and processing time t_j (travel time to reach it), minimising $\sum p_j a_j$ is equivalent to scheduling jobs in order of decreasing p_j/t_j (the *Smith's Rule* heuristic from single-machine scheduling). In routing, however, the “processing times” are not fixed because travel times depend on the order of visits, making the problem combinatorially harder.

14.4 — Commercial VRPs vs. Disaster Relief VRPs I

The authors devote a section to carefully distinguishing commercial VRPs from disaster relief VRPs, because researchers often adapt commercial models to humanitarian settings without fully accounting for the structural differences. Understanding these differences is essential for building models that actually help in practice.

14.4 — Commercial VRPs vs. Disaster Relief VRPs II

Commercial VRP vs. Disaster Relief VRP: Key Differences

Dimension	Commercial VRP	Disaster Relief VRP
Primary Objective	Minimise cost / distance	Minimise suffering / response time
Fleet Availability	Known, controlled	Uncertain, may be damaged
Demand	Known or forecast	Unknown / rapidly evolving
Road Network	Reliable	May be blocked or destroyed
Time Horizon	Planned in advance	Real-time, dynamic re-routing
Objectives	Single (cost)	Multiple (equity, speed, coverage)
Vehicle Type	Homogeneous fleets	Diverse (trucks, helicopters, boats)
Priority	Not typically considered	Life-saving priority is critical
Return to Depot	Required each trip	May not return; one-way routes
Uncertainty	Low	Extremely high

14.4.1 — Low Cost vs. Low Arrival Time I

In commercial logistics, the dominant objective is to **minimise total operating cost**: fuel, driver wages, vehicle depreciation. Distance is a proxy for cost, so classical VRPs minimise total route length.

In disaster relief, the primary objective is to **minimise arrival time** (response time) or to **maximise welfare** of affected populations. Cost matters, but it is subordinate to urgency.

A Fundamental Tension: Distance vs. Arrival Time

The route that minimises *total* travel distance is not necessarily the route that minimises *maximum* arrival time. Consider a simple example with one vehicle and two locations A and B:

- $d(\text{depot}, A) = 10$, $d(A, B) = 5$, $d(B, \text{depot}) = 12$
- Route (depot $\rightarrow$ A $\rightarrow$ B $\rightarrow$ depot): total distance = 27, arrival at A = 10, at B = 15.
- Route (depot $\rightarrow$ B $\rightarrow$ A $\rightarrow$ depot): total distance = 27, arrival at B = 12, at A = 17.

Both routes have the same total distance, but the first gives location A an earlier arrival (10 vs. 17). If A is more urgent (e.g., a medical emergency), the first route is unambiguously better under a humanitarian objective, even though the distances are identical.

14.4.1 — Low Cost vs. Low Arrival Time II

The mathematical implication is profound. Commercial VRP objectives are **symmetric in customer order**: the total distance of a route depends only on which customers are on the route, not on the order in which they are visited (given a fixed set of customers, TSP-optimal ordering minimises total distance).

Disaster relief objectives are **order-sensitive**: the same set of locations, visited in a different order, can give dramatically different arrival times and thus different humanitarian impact.

Key Insight

This order-sensitivity means that **classical VRP solvers, which optimise insertion order primarily for distance, are poorly suited to disaster relief without modification**. Algorithms must explicitly account for arrival-time objectives and priority constraints.

Campbell, VanderBerg, and Werneck (2008) formalised this observation by studying the **deliveries with time windows** variant of disaster VRP, showing that even a small change in arrival-time objective leads to substantially different optimal routes.

14.4.2 — Multiple Objectives and Fleet Considerations I

Commercial VRPs almost always optimise a single objective (usually cost or distance). Even when multiple objectives are mentioned, they are typically reduced to a single weighted sum.

Disaster relief VRPs inherently involve multiple, often conflicting objectives:

- 1 Minimise total distance (resource efficiency)
- 2 Minimise maximum arrival time (urgency)
- 3 Maximise coverage (reach)
- 4 Maximise equity (fairness)
- 5 Minimise unsatisfied demand (total welfare)

14.4.2 — Multiple Objectives and Fleet Considerations II

These objectives cannot generally be reduced to a single scalar without losing important information. Multi-objective optimisation methods — including **Pareto frontier** analysis and **goal programming** — are therefore more appropriate.

Pareto Optimality in Disaster Routing

A solution x^* is **Pareto optimal** if there is no other feasible solution x that is at least as good on all objectives and strictly better on at least one. The set of all Pareto-optimal solutions is called the **Pareto frontier** (or Pareto front). Decision-makers can then choose from the Pareto frontier based on their preferences, rather than having a single “optimal” answer imposed on them.

14.4.2 — Multiple Objectives and Fleet Considerations III

Fleet heterogeneity is another key difference. Commercial fleets are typically homogeneous: all vehicles have the same capacity and speed. Disaster relief may require a mix of:

- **Trucks:** High capacity, limited to passable roads.
- **Helicopters:** Can reach inaccessible areas but have lower capacity and high operating costs.
- **Boats:** Required when roads are flooded.
- **Pack animals:** Used in remote mountain terrain.

This heterogeneity requires a **mixed fleet VRP**, where each vehicle type has different parameters (Q_k, v_k, c_k) : capacity Q_k (Q sub k), speed v_k (v sub k), cost per unit distance c_k (c sub k).

Additionally, vehicle availability itself may be uncertain: a truck may break down, a helicopter may be requisitioned by the military. Robust routing models account for this by building contingency plans or by formulating stochastic programmes over vehicle availability scenarios.

14.4.3 — The Min-Max Tour Length Problem I

One particularly important problem that arises in disaster relief (but rarely in commercial logistics) is the **Min-Max VRP**, also called the **Minimax VRP**: minimise the length of the longest route among all vehicles.

This objective emerges naturally in two situations:

- 1 **Time-to-completion**: If all vehicles must finish before the next re-supply cycle, the cycle time is determined by the *slowest* vehicle (the longest route). Minimising the longest route minimises the total cycle time.
- 2 **Workload balance**: Fairness to drivers or logistics teams requires that no single vehicle is dramatically overloaded relative to others.

14.4.3 — The Min-Max Tour Length Problem II

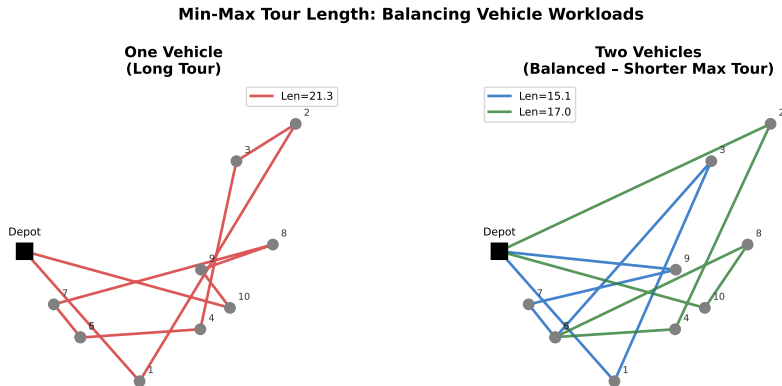


Figure: Left: One vehicle completes a very long tour. Right: Two balanced vehicles each complete shorter tours. The min-max objective drives the planner to choose the balanced solution even if it uses more total distance.

14.4.3 — The Min-Max Tour Length Problem III

Formal Min-Max VRP Model

Let K be the number of vehicles and R_k (R sub k) the route assigned to vehicle k . Let $L(R_k)$ denote the length (total distance or time) of route R_k . The **Min-Max VRP** minimises:

$$\min \max_{k=1,\dots,K} L(R_k)$$

subject to standard VRP constraints (all demand locations visited exactly once, capacity constraints, etc.).

Why This is Harder Than Minimising Total Distance

The min-sum objective (total distance) is a linear function of routing decisions, which makes it well-suited for linear programming relaxations. The min-max objective is a *nonlinear* (piecewise linear) function: it equals the largest of the route lengths. This can be linearised by introducing an auxiliary variable $z \geq L(R_k)$ for all k and minimising z , but it substantially increases the complexity of the model.

14.4.3 — The Min-Max Tour Length Problem IV

Empirical results from Borràs and Pastor (2002) and Huang, Smilowitz, and Balcik (2012) show that **min-max solutions can be 15–30% longer in total distance** than min-sum solutions, but they reduce the maximum route length (and thus the total operation time) by 20–40%. In disaster relief, this trade-off is typically worth making.

14.5 — VRP Models for Disaster Relief: Overview I

The chapter surveys two main categories of VRP models for disaster relief:

- 1 **Distribution of relief supplies:** Getting food, water, medicine, and shelter materials from depots to affected communities.
- 2 **Evacuation routing:** Moving vulnerable people (injured, elderly, disabled) from danger zones to safety.

Within each category, models range from classical VRP extensions (adding priority or equity constraints to the standard CVRP) to highly specialised formulations designed for specific disaster types.

14.5 — VRP Models for Disaster Relief: Overview II

Common Modelling Elements

Across all disaster relief VRP models, the following elements appear repeatedly:

- **Graph** $G = (V, E)$ where V is the set of nodes (depot + demand locations) and E is the set of edges (road segments).
- **Vehicle fleet** $K = \{1, \dots, m\}$ with capacities Q_k and speeds v_k .
- **Demand** d_j at each location $j \in V \setminus \{0\}$.
- **Priority** p_j or urgency weight w_j for each demand location.
- **Time windows** $[e_j, l_j]$ (earliest and latest acceptable arrival times) in some models.

14.5.1 — Distribution of Relief Supplies: Demand Satisfaction Models I

The most fundamental disaster relief routing problem asks: **given limited vehicle capacity and a damaged road network, how should vehicles route to deliver the maximum possible supplies to affected communities?**

14.5.1 — Distribution of Relief Supplies: Demand Satisfaction Models II

Basic Capacitated VRP for Relief Distribution

Let the node set be $V = \{0, 1, \dots, n\}$ where node 0 is the depot and nodes $1, \dots, n$ are demand locations. Let $x_{ijk} \in \{0, 1\}$ (x sub i j k) = 1 if vehicle k travels directly from node i to node j . Let $y_{jk} \in \{0, 1\}$ (y sub j k) = 1 if location j is served by vehicle k . Let $s_{jk} \geq 0$ (s sub j k) be the quantity delivered to location j by vehicle k .

Objective: Maximise total demand satisfaction:

$$\max \sum_{k \in K} \sum_{j=1}^n s_{jk}$$

Subject to:

$$\sum_{k \in K} \sum_{j=1}^n s_{jk} \leq D \quad (\text{total supply available } D) \quad (1)$$

$$\sum_{k \in K} s_{jk} \leq d_j \quad \forall j \quad (\text{do not exceed demand}) \quad (2)$$

$$\sum_{k \in K} x_{ijk} = y_{jk} \quad \forall j \in \{1, \dots, n\}$$

Team Orienteering Problem for Relief Distribution I

When not all demand locations can be served within the available time or vehicle capacity, the problem becomes one of **selective routing**: choose *which* locations to visit (not just in what order).

This maps to the **Team Orienteering Problem (TOP)**, sometimes called the **Multiple Depot Team Orienteering Problem (MDTOP)**. The TOP is an extension of the Orienteering Problem, itself derived from the sport of orienteering where competitors choose which checkpoints to visit to maximise collected points.

Team Orienteering Problem for Relief Distribution II

Team Orienteering Problem (TOP) Formulation

Given:

- A set V of nodes (locations), each with a reward $r_j \geq 0$,
- m vehicles (“teams”), each with a maximum route length (or time budget) $T_{\max}$,
- A start depot and end depot (possibly the same).

Find at most m routes — one per vehicle — such that:

$$\max \sum_{k=1}^m \sum_{j \in V} r_j \cdot y_{jk}$$

subject to: each route has total length $\leq T_{\max}$, each location is visited at most once across all routes, and routes start and end at the designated depots.

Team Orienteering Problem for Relief Distribution III

In disaster relief, the **reward** r_j at location j is typically interpreted as the **demand** d_j (the benefit of delivering to that location), or as a **priority score** combining urgency and demand:

$$r_j = p_j \cdot d_j$$

where p_j (p sub j) is the priority of the location and d_j (d sub j) is its demand. Locations with high demand *and* high urgency get the highest reward and are thus most likely to be selected.

Team Orienteering Problem for Relief Distribution IV

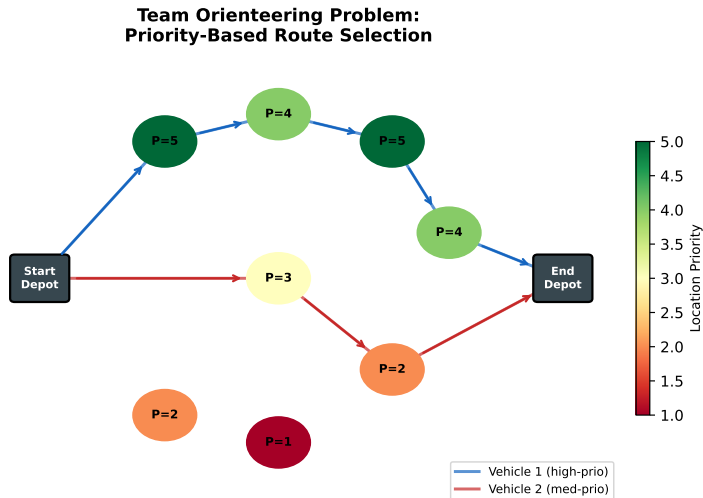


Figure: Team Orienteering Problem routing. High-priority locations (green/yellow) are visited first. Low-priority locations (red) may be skipped if the time budget does not allow their inclusion. The colour bar on the right^{63 / 88}

Demand Satisfaction with Route Duration Constraints I

Raths and Lorenz (2012) study a disaster relief VRP where:

- Vehicles have both capacity constraints *and* route duration constraints (a maximum time per route).
- Not all demand needs to be satisfied (partial delivery allowed).
- The objective is to **maximise total demand satisfaction** subject to feasibility.

Raths–Lorenz Model

Let f_j (f sub j) be the fraction of demand satisfied at location j ($0 \leq f_j \leq 1$). The model maximises:

$$\max \sum_{j=1}^n w_j \cdot f_j \cdot d_j$$

where w_j (w sub j) is the weight (importance) of location j , f_j is the fraction delivered, and d_j is the total demand. This is equivalent to maximising *weighted total units delivered*.

The key constraint is that routes must be completed within a maximum duration T (e.g., 8 hours per vehicle per day), which limits both the number of locations visited and the total quantity delivered.

Demand Satisfaction with Route Duration Constraints II

to solutions with equal, but equally bad, service levels. This type of solution is shown in Figure 14.12. Here, two vehicles traverse the same route in opposite directions and deliver one unit of supply. The total demand of each customer (two units) is satisfied after time tt , where tt is the travel time of each vehicle (we assume a symmetric distance matrix). This solution minimizes the objective function at the cost of the response time.

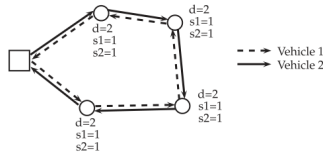


Figure 14.12. *Equal but inefficient delivery when demand satisfaction may be split. Demand (d) is given for each location. The supply $s1$ is delivered by vehicle 1 and supply $s2$ is delivered by vehicle 2 [21].*

The authors propose an objective function that promotes partial deliveries to locations in order to save goods for delivery to other destinations. However, in contrast to the previously described objective, this one favors a rapid delivery through the use of a convex penalty term that depends on the fraction of unsatisfied demand at each location. The lower the satisfaction rate, the higher the penalty; i.e., the penalty decreases with each unit delivered. The objective is to minimize the sum of the penalty costs that are incurred over time. Initial differences in the needs are not considered.

Equity-Based Distribution Models I

Several papers explicitly model **equity** as a primary objective, arguing that in humanitarian contexts, fair distribution is a moral imperative, not just a performance metric.

Huang, Smilowitz, and Balcik (2012): Routing with Fairness

This paper introduces a VRP where the objective is to minimise the maximum **deprivation cost** $\phi_j(a_j)$ (phi sub j of a sub j) across all demand locations. The deprivation cost is a convex, increasing function of arrival time: the longer a community waits, the greater its suffering.

Specifically:

$$\min \max_j \phi_j(a_j)$$

Here $\phi_j(a_j)$ (phi: the Greek letter for a cost function) is a location-specific cost function. The minimax structure ensures the *worst-served* community is improved first, embodying a Rawlsian fairness principle.

How to Read This Formula

The formula says: optimise for the community that suffers the most. Unlike minimising total cost (which could ignore one very poor community to serve many comfortable ones), minimax forces the solution to “raise the floor” — improve the worst outcome. This is the humanitarian version of the social welfare principle.

Equity-Based Distribution Models III

to operate economically in order to be able to provide assistance until self-sustainability of the population is recovered.

Commercial and relief routing models are very similar in terms of cost as a performance metric. Relief routing models consider travel costs with lower priority or in conjunction with humanitarian constraints. The first priority in Bish [8] and Huang, Smilowitz, and Balcik [21] is the optimization of humanitarian oriented metrics. Routing costs are only considered to break ties if the primary objective has multiple optimal solutions. In Balcik, Beamon, and Smilowitz [3] and Lin et al. [33], humanitarian objectives and routing costs are aggregated into one objective function. The respective weights are set to favor the humanitarian metrics. Minimizing the travel distance is the only objective in Panchamgam et al. [47]. The urgency of need at locations is taken into account by precedence constraints in the sequence of visits. The objective function in Wohlgemuth, Oloruntoba, and Clausen [69] minimizes the number of vehicles used and the total travel time. The applied dynamic solution approach incorporates two features of disaster relief operations: changing road conditions (i.e., varying travel times) and the arrival of new orders during the execution of the tours.

In the routing and allocation problem presented in Hodgson, Laporte, and Semet [19], transportation costs are minimized while the walking distance from the unvisited locations to their nearest tour stop is constrained. In Doerner, Focke, and Gutjahr [15], Jozefowicz, Semet, and Talbi [26], Nolz et al. [38], Nolz, Semet, and Doerner [40], Nolz, Doerner, and Hartl [39], and Tricoire, Graf, and Gutjahr [61], a multi-objective optimization approach is applied where costs are optimized along with the walking distance or the share of the population that is covered within a specified distance.

14.5.2 — Evacuation Routing I

Evacuation routing is the second major category of disaster relief VRP. The goal is to move people — not goods — from danger zones to safety, using a fleet of vehicles operating on a (possibly damaged) road network.

Evacuation routing has several unique features that distinguish it from supply distribution:

- ❶ **The “commodity” is human:** Vehicles transport people rather than goods, imposing safety and dignity constraints.
- ❷ **One-way flow:** Vehicles go from hazard zones to shelters; the return trip brings no immediate benefit.
- ❸ **Contraflow operations:** Road capacity can be doubled by reversing normally inbound lanes to carry outbound evacuation traffic.
- ❹ **Capacity of shelters:** Shelters have limited capacity; routing must account for where evacuees can actually go.
- ❺ **Self-evacuation:** Many people evacuate independently by car, creating massive traffic congestion that affects vehicle routing.

14.5.2 — Evacuation Routing II

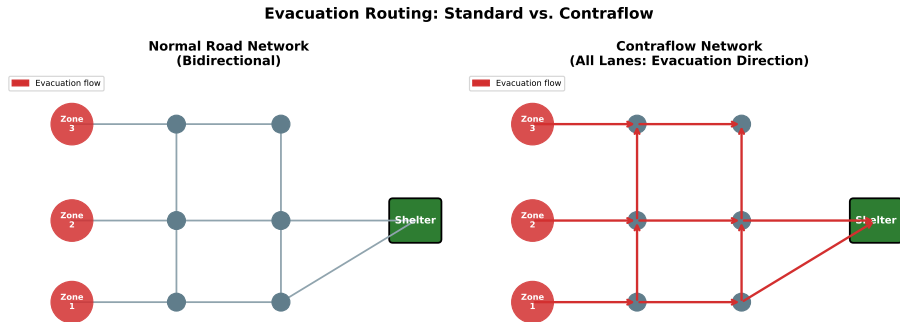


Figure: Left: Standard bidirectional road network. Right: Contraflow configuration where all lanes are re-directed outbound to maximise evacuation throughput. Contraflow effectively doubles the capacity of evacuation routes but requires coordination with traffic management authorities.

14.5.2 — Evacuation Routing III

Network Structure for Evacuation

Evacuation is modelled on a directed graph $G = (V, A)$ where V is the set of nodes (evacuation zones + shelters + road intersections) and A is the set of directed arcs (road segments with a specific direction of travel). **Contraflow** is modelled by adding reverse arcs to normally one-directional road segments, with appropriate capacity and travel time parameters.

Yi and Özdamar (2007): Dynamic Logistics Coordination

Yi and Özdamar model evacuation and supply distribution *simultaneously* in a multi-period network flow problem. In period t , vehicles can carry either supplies (inbound to disaster zones) or evacuees (outbound to shelters), subject to vehicle capacity constraints.

This is a **dynamic network flow problem**:

- Nodes represent (location, time) pairs.
- Arcs represent vehicle movements or waiting in place.
- Flow on each arc is bounded by vehicle capacity.

The objective balances the urgency of delivering supplies against the urgency of evacuating injured people.

Evacuation with Heterogeneous Demand

Not all evacuees have the same mobility. The **special needs** population (elderly, disabled, hospitalised) cannot self-evacuate and must be given priority access to vehicles. Ozbay and Yazici (2006) study the allocation of transit buses to special-needs evacuation, formulating it as a VRP where:

- Each bus picks up evacuees from multiple pickup points (nursing homes, hospitals) and delivers them to a shelter.
- Bus routes are constrained by travel time and vehicle capacity.
- The objective is to minimise the number of buses required to evacuate all special-needs individuals within a time window.

Evacuation Routing: Key Models III

Contraflow Optimisation (Lim, 2008)

Lim models contraflow assignment as an optimisation problem: given a set of road segments that *could* be reversed, decide which segments to reverse (and at what cost) in order to maximise network throughput for evacuation.

Formally, let $b_e \in \{0, 1\}$ ($b \text{ sub } e$) = 1 if arc e is reversed (contraflow), 0 otherwise. The total contraflow capacity on the evacuation corridors is:

$$\text{Throughput} = \sum_{e \in E} (c_e + b_e \cdot c'_e)$$

where c_e is the original capacity of arc e and c'_e ($c \text{ prime sub } e$) is the additional capacity gained by reversing it. The constraint is that implementing contraflow on certain segments conflicts with implementing it on adjacent segments (you cannot reverse roads in both directions at the same intersection).

Key Insight: Evacuation is a Supply Chain in Reverse

While supply distribution moves goods *toward* affected populations, evacuation moves people *away* from danger. The vehicle routing mathematics is similar, but the priorities, constraints, and coordination requirements are different. Effective evacuation planning requires routing models to be integrated with traffic management, shelter management, and public communication systems.

Dynamic Routing in Disaster Settings I

A major distinguishing feature of disaster relief VRP is **dynamism**: both the environment and the information available to planners change rapidly during an operation.

Dynamic VRP (DVRP) Formulation

In a DVRP, the input data $\{d_j, p_j, t_{ij}\}$ (demand, priority, travel time) is not fully known at the start of the planning horizon. New information arrives at discrete time points $\tau_1, \tau_2, \dots$ (τ : tau, the Greek letter for time steps). At each τ_t , the routing plan is **re-optimised** based on all available information.

The degree of dynamism δ (delta, the Greek letter) is defined as:

$$\delta = \frac{\text{Number of requests known only after start}}{\text{Total number of requests}}$$

Commercial DVRP typically has $\delta \approx 0.1\text{--}0.3$; disaster relief DVRP can have $\delta \approx 0.7\text{--}1.0$, meaning most demand is unknown at the start of operations.

Dynamic Routing in Disaster Settings II

Re-optimisation Strategies

When new information arrives (a new affected area is reported, a road is found to be blocked), the dispatcher must decide:

- 1 **Myopic re-routing:** Immediately update routes based on current information, ignoring future uncertainty. Simple and fast, but may make poor long-term decisions.
- 2 **Anticipatory re-routing:** Use probabilistic models of where future requests are likely to arise and build routes that are robust to likely future scenarios. More complex but can reduce total response time by 10–20%.
- 3 **Scenario-based robust routing:** Enumerate a set of likely future scenarios and find a routing plan that performs well across all of them (a “hedged” plan).

Wohlgemuth, Oloruntoba, and Clausen (2012) study **dynamic vehicle routing with anticipation** in disaster relief, showing that anticipatory strategies outperform myopic re-routing when the disaster evolves predictably (e.g., a flood that advances in a known direction).

Key Insight

The fundamental challenge of dynamic disaster routing is **balancing responsiveness to new information against the disruption cost of re-routing vehicles in the field**. Constant re-routing may confuse field teams and increase travel distances; too little re-routing leads to poor adaptation.

Python Implementation: Priority-Weighted Greedy Routing

The following Python code implements a **priority-weighted nearest-neighbour heuristic** for disaster relief routing. The heuristic builds a route by always selecting the unvisited location with the best priority-to-distance ratio.

```
import numpy as np

def priority_nn_heuristic(depot, locations, demands, priorities,
                          vehicle_capacity):
    """
    Priority-weighted nearest-neighbour heuristic for disaster VRP.

    Parameters
    -----
    depot      : (x, y) coordinates of the depot
    locations   : list of (x, y) coordinates of demand locations
    demands     : list of demand values d_j for each location
    priorities  : list of priority weights p_j for each location
    vehicle_capacity : Q (maximum load per vehicle)

    Returns
    -----
    routes : list of routes, each route is a list of location indices
    """
    n = len(locations)
    unvisited = set(range(n))    # indices of unvisited locations
    routes = []                  # list of completed routes
```

Python Implementation: Computing Arrival Times and Equity Metrics

```
import numpy as np

def route_arrival_times(depot, locations, route, speed=1.0):
    """
    Compute arrival times at each stop in a route.

    Parameters
    -----
    depot      : (x, y) tuple -- starting point
    locations   : list of (x, y) tuples for all demand nodes
    route       : list of location indices in visit order
    speed       : vehicle speed (distance units per time unit)

    Returns
    -----
    arrivals   : list of arrival times (one per stop in route)
    """
    arrivals = []
    current = np.array(depot)
    time = 0.0
    for j in route:
        dest = np.array(locations[j])
        dist = np.linalg.norm(dest - current)
        time += dist / speed
        arrivals.append(time)
        current = dest
    return arrivals
```

14.6 — Conclusions and Future Research Directions I

The authors conclude by summarising the state of the disaster relief VRP literature and identifying the most important open research questions.

Main Conclusions

- 1 **The field is young but growing rapidly.** The number of papers on humanitarian logistics and disaster relief routing has approximately doubled every five years since 2000.
- 2 **Commercial VRP models require substantial modification.** Simply applying existing VRP heuristics to disaster data produces plans that may be efficient in distance but poor in humanitarian terms.
- 3 **Multi-objective formulations are essential.** Capturing the trade-offs between speed, equity, and coverage requires genuinely multi-objective optimisation, not just a weighted single objective.
- 4 **Vehicle routing compatibility matters.** Requires that all locations be reachable by at least one vehicle type, even if roads are blocked.
- 5 **Dynamic and stochastic elements are unavoidable.** Static, deterministic models miss the most important operational challenge: responding to rapidly changing information.

14.6 — Conclusions and Future Research Directions II

Open Research Directions

The authors identify several areas where future research is most needed:

- ➊ **Pre-positioning with network uncertainty.** Better models that account for the possibility that specific roads or depots will be unavailable after a disaster, requiring robust pre-positioning decisions.
- ➋ **Integration across phases.** Models that span the preparedness–response–recovery continuum, rather than treating each phase in isolation.
- ➌ **Coordination among multiple agencies.** Mechanisms for coordinating routing decisions across NGOs, government agencies, and military organisations that may have conflicting objectives.
- ➍ **Incorporating self-evacuation behaviour.** Realistic evacuation models must account for the fact that most people evacuate independently, creating congestion that affects official vehicle routing.
- ➎ **Algorithmic scalability.** Disaster-scale instances can involve thousands of locations and dozens of vehicles. Exact methods are too slow; better large-scale heuristics and metaheuristics are needed.

14.6 — Conclusions and Future Research Directions III

Specific Research Gap: Vehicle Routing Compatibility

One requirement that the authors flag as insufficiently studied is the concept of **routing compatibility** in disaster VRP. In commercial settings, we typically assume that all vehicles can reach all customers. In disaster settings, this assumption fails: a heavy truck cannot traverse a washed-out bridge; a helicopter cannot land in a dense urban canyon.

Future models should explicitly encode an **access compatibility matrix** $A_{kj} \in \{0, 1\}$ ($A_{kj} = 1$ if vehicle type k can access location j , 0 otherwise). This matrix, combined with routing constraints, forces the model to assign each location to a compatible vehicle type.

Final Key Insight

Disaster relief routing is one of the most socially impactful applications of combinatorial optimisation. Even modest improvements in routing efficiency — reducing average response time by 20% or improving supply equity by 30% — can translate directly into lives saved. This makes it both a compelling research area and a responsibility for the operations research community.

Summary: Key Concepts and Models I

This frame provides a consolidated reference for the main concepts and formulas introduced in this chapter.

Performance Metrics:

- **Arrival time** at location j : $a_j = \sum_{\ell=1}^m t_{i_{\ell-1}, i_{\ell}}$ (cumulative travel time along the route to location j).
- **Min-max response time**: $\min \max_j a_j$ (minimise the latest arrival).
- **Min-sum response time**: $\min \sum_j a_j$ (minimise total waiting time).
- **Demand satisfaction rate**: $\rho_j = s_j/d_j$ (rho sub j).
- **Service equity**: $\min \text{Var}(\rho_j)$ or $\max \min_j \rho_j$.

VRP Variants for Disaster Relief:

- **Priority-weighted VRP**: Objective $\min \sum_j p_j a_j$.
- **Team Orienteering Problem (TOP)**: Select which locations to visit within a time budget to maximise reward.
- **Min-Max VRP**: $\min \max_k L(R_k)$ (balance workloads).
- **Dynamic VRP (DVRP)**: Real-time re-optimisation as information evolves.
- **Stochastic VRP**: Demand d_j is a random variable; plan under uncertainty.

Summary: Key Concepts and Models II

Commercial vs. Disaster Relief VRP — At a Glance:

Dimension	Commercial VRP	Disaster Relief VRP
Objective	Min cost / distance	Min time; Max equity
Fleet	Homogeneous	Heterogeneous
Demand	Known	Uncertain
Network	Reliable	Disrupted
Time horizon	Planned	Real-time
Priorities	Absent	Life-critical
Return to depot	Required	May not return

Summary: Key Concepts and Models III

Take-Home Message

Vehicle routing in disaster relief is not simply “standard VRP with a different dataset.” The objectives, constraints, and operational environment are fundamentally different. Effective humanitarian logistics requires purpose-built VRP models that encode urgency, equity, uncertainty, and multi-vehicle coordination — and algorithms fast enough to be useful in the hours immediately following a disaster.

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